

Algebra I

Հանրահաշիվ I

COURSE CODE	MTH-A1
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Typically Grade 8 or 9
DURATION	108–126 instructional hours (approx. 90 lessons)
PREREQUISITE	Grade 8 Mathematics or Pre-Algebra, or placement by assessment: multi-step equations, slope, function notation.
LEADS TO	Geometry (MTH-GE) — or Algebra II (MTH-A2) on an integrated pathway

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Algebra I is the gateway course of American secondary mathematics. It establishes the function as the central object of study and develops three families in depth: linear, exponential, and quadratic. Students learn to see algebraic structure — to recognize that an expression can be rewritten to reveal something that was not previously visible — and to move fluently among symbolic, graphical, numerical, and verbal representations. The course closes with statistics for bivariate data, including linear regression and the distinction between correlation and causation.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Algebra I is the course in which the largest number of American students first decide that they are 'not a math person'. Վարժարան treats that as a curricular failure rather than a student failure, and builds the course accordingly. Three commitments follow. First, every symbolic technique is anchored to a graphical or contextual meaning, so that a student who forgets a procedure can rebuild it. Second, modelling is not an end-of-chapter application but a recurring thread — students are asked from the first unit to decide which function family fits a situation. Third, the teacher's response to a wrong answer is always to locate the misconception, never to repeat the procedure louder.

PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves multi-step linear equations with variables on both sides, with checking	The computational floor of the entire course; remediate before Unit 2 if weak.
Computes slope from any representation and interprets it with units	Predicts success in Unit 3; a very common gap.
Uses function notation and applies the definition of a function	If absent, Unit 1 must be extended; do not proceed to Unit 3 without it.
Computes fluently with negative rational numbers and exponents	Sign and exponent errors are the largest cause of lost accuracy in quadratics.
Translates a described situation into an equation	Modelling runs through every unit; weakness here surfaces as 'I don't know how to start'.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Interpret the structure of expressions and rewrite them to reveal properties of the quantities represented.
- 2 Create equations and inequalities in one or more variables and use them to solve problems and model relationships.
- 3 Understand solving equations as a process of reasoning and explain each step by a justified property.
- 4 Solve linear equations and inequalities, and systems of linear equations, and represent solution sets graphically.
- 5 Understand the concept of a function, use function notation, and interpret functions arising in applications.
- 6 Analyze linear, exponential, and quadratic functions using different representations.
- 7 Build a function that models a relationship between two quantities and build new functions from existing ones.
- 8 Construct and compare linear and exponential models and interpret their parameters in context.
- 9 Perform arithmetic on polynomials and understand the closure of polynomials under these operations.
- 10 Factor quadratic expressions and use the factored form to reveal zeros.
- 11 Solve quadratic equations by inspection, factoring, completing the square, the quadratic formula, and graphing.
- 12 Understand and apply the properties of exponents to radical and rational exponent expressions.
- 13 Summarize, represent, and interpret data on two categorical and quantitative variables.
- 14 Interpret linear models, distinguish correlation from causation, and assess the fit of a model.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 12-13 LESSONS

Expressions, Equations, and Reasoning

Արտահայտություններ, հավասարումներ և տրամաբանում

TOPICS COVERED

- Quantities, units, and the choice of an appropriate scale in modelling
- Interpreting the parts of an expression: terms, factors, coefficients
- Interpreting a complicated expression by viewing parts as a single entity
- Properties of operations as the justification for algebraic manipulation
- Solving linear equations as a chain of justified steps
- Equations with no solution and with infinitely many solutions
- Literal equations and rearranging formulas for a specified variable
- Linear inequalities in one variable and their solution sets
- Compound inequalities: and, or
- Absolute value equations and inequalities
- Creating equations and inequalities from context
- Interpreting the solution in the context of the modelling constraints

LEARNING OBJECTIVES

- 1 Justify each step in solving an equation by naming the property applied.
- 2 Rearrange a formula to isolate any specified variable.
- 3 Solve absolute value equations and inequalities and explain the two cases using the distance definition.
- 4 Create an equation or inequality from a described situation and interpret the solution set in context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves linear equations and inequalities at 92% accuracy with justification available on request.
- Rearranges literal equations correctly in 8 of 10 trials.
- Solves absolute value inequalities correctly, including the direction of the compound statement, in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Absolute value inequalities are solved by dropping the bars and adding $\pm$.	Return to the distance definition: $ x - 3 < 5$ asks for numbers within 5 of 3. Draw it on a number line before writing any compound inequality.
Literal equations are treated as a different skill from numerical equations.	They are the same operations with letters. Solve $3x + 5 = 20$ and $ax + b = c$ side by side, step for step.

MISCONCEPTION	RECOMMENDED RESPONSE
The inequality direction is not reversed when multiplying by a negative variable expression.	Multiplying by a variable of unknown sign is not permitted at all. Name this explicitly — it prevents a class of errors that persists into Algebra II.

TEACHING NOTE

The habit of naming the property that justifies each step feels slow in this unit and pays for itself for the rest of the student's mathematical education. It converts algebra from manipulation into argument.

UNIT 2 · 10-11 LESSONS**Functions and Their Representations**

Ֆունկցիաներ և դրանց ներկայացումները

TOPICS COVERED

- The function definition, domain, and range
- Function notation, evaluation, and solving $f(x) = c$
- Domain in context versus domain of the algebraic rule
- Reading key features from a graph: intercepts, intervals of increase and decrease, maxima and minima, end behaviour
- Average rate of change over an interval
- Relating the domain to the quantitative relationship it describes
- Sketching a graph from a verbal description of key features
- Comparing properties of two functions given in different representations
- Piecewise-defined functions and step functions
- Recursive and explicit definitions of sequences
- Sequences as functions on the integers

LEARNING OBJECTIVES

- 1 Interpret key features of a graph in the context the function models.
- 2 Compute an average rate of change over an interval and interpret it with units.
- 3 Determine an appropriate domain for a function used as a model, and justify it.
- 4 Write both a recursive and an explicit rule for an arithmetic sequence.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Reads and interprets all key graph features correctly in 9 of 10 trials.
- Computes and interprets average rate of change in 9 of 10 trials with units.
- Writes recursive and explicit rules for a sequence in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The graph of a function is read as a picture of a physical path rather than as a relationship.	The classic 'graph as picture' error. Use a distance-time graph for a person walking back and forth and require the student to narrate it.

MISCONCEPTION	RECOMMENDED RESPONSE
Domain is stated as the algebraic domain even when the context forbids it.	A model for the number of tickets sold has a domain of non-negative integers. Always ask what the input <i>means</i> before stating the domain.
Average rate of change is confused with the value of the function.	It is a slope between two points, not a height. Compute both for the same interval and contrast them explicitly.

TEACHING NOTE

This unit is representation-heavy by design. A student who can only work symbolically has not learned Algebra I; the ability to move among four representations is the actual outcome.

UNIT 3 • 13-14 LESSONS**Linear Functions, Systems, and Modelling**

Գծային ֆունկցիաներ, համակարգեր և մոդելավորում

TOPICS COVERED

- Linear functions as constant rate of change
- Slope-intercept, point-slope, and standard form, and when each is convenient
- Writing a linear function from two points, a graph, a table, or a description
- Parallel and perpendicular lines and their slopes
- Transformations of the parent function $f(x) = x$: vertical shifts and stretches
- Solving systems by graphing, substitution, and elimination
- Choosing an efficient method
- Systems with no solution and infinitely many solutions
- Graphing linear inequalities in two variables
- Systems of linear inequalities and feasible regions
- Modelling with systems, including constraint problems
- Interpreting the parameters of a linear model in context
- Arithmetic sequences as linear functions on the integers

LEARNING OBJECTIVES

- 1 Write the equation of a line from any adequate description and convert among the three forms.
- 2 Solve a system by an efficiently chosen method and justify the choice.
- 3 Graph a system of linear inequalities and interpret the feasible region in context.
- 4 Build a linear model from a described situation and interpret both parameters with units.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Writes and converts linear equations at 92% accuracy.
- Solves systems at 92% accuracy across all three methods.
- Interprets model parameters in context with units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Perpendicular slopes are taken as the negative, or as the reciprocal, but not both.	The product of the slopes is -1 . Verify with a graph on a square grid — a non-square window makes perpendicularity look wrong and reinforces the error.
The wrong region is shaded for an inequality.	Test a point, always. The origin is usually the easiest test point, and testing is faster than remembering a rule.
The solution of a system is reported without checking in both equations.	A solution must satisfy both. Require the double check; it catches nearly all elimination errors.

TEACHING NOTE

Feasible-region problems are the student's first encounter with optimization under constraints and are worth real time. They also make an abstract topic visibly useful, which matters at this stage.

UNIT 4 • 11-12 LESSONS**Exponential Functions and Sequences**

Ֆունկցիաների և հաջորդականությունների

TOPICS COVERED

- Properties of integer exponents, reviewed and extended
- Rational exponents and the connection to radicals
- Simplifying radical expressions
- Exponential growth and decay as repeated multiplication
- The general form $f(x) = a \cdot b^x$ and the meaning of a and b
- Recognizing exponential behaviour in a table by constant ratios
- Comparing linear and exponential growth over the long run
- Percent growth and decay; compound interest
- Geometric sequences and their explicit and recursive forms
- Graphs of exponential functions: intercepts, asymptotes, end behaviour
- Transformations of exponential functions
- Choosing between a linear and an exponential model from data
- Solving simple exponential equations by rewriting with a common base

LEARNING OBJECTIVES

- 1 Distinguish linear from exponential behaviour in a table, a graph, and a context.
- 2 Interpret a and b in $f(x) = a \cdot b^x$ in a modelling context with units.
- 3 Convert between radical and rational exponent notation and simplify.
- 4 Justify why an exponential quantity eventually exceeds any linear one.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Classifies data as linear or exponential correctly in 9 of 10 trials with justification.
- Interprets exponential model parameters in context in 9 of 10 trials.
- Simplifies rational-exponent and radical expressions at 88% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Exponential growth is described as 'fast' rather than as multiplicative.	An exponential function with $b = 1.01$ grows slowly at first and still overtakes any linear function. Graph them together over a long domain.
The base b and the growth rate r are confused: 5% growth entered as $b = 0.05$.	$b = 1 + r$. Write both quantities explicitly in every model and state what each means.
$(a + b)^n$ is expanded as $a^n + b^n$.	Test with numbers: $(2 + 3)^2$ is 25, not 13. One counterexample, produced by the student, is worth many corrections.

TEACHING NOTE

The comparison of long-run linear and exponential behaviour is one of the most consequential ideas a secondary student meets, well beyond mathematics. Spend a full lesson on it with real data.

UNIT 5 • 11-12 LESSONS

Polynomials and Factoring

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TOPICS COVERED

- Polynomial vocabulary: degree, leading coefficient, standard form
- Adding and subtracting polynomials; closure
- Multiplying polynomials and the area model
- Special products: difference of squares and perfect square trinomials
- Factoring out the greatest common factor
- Factoring $x^2 + bx + c$
- Factoring $ax^2 + bx + c$
- Factoring special forms
- Factoring by grouping
- Choosing a factoring strategy
- The zero product property
- Solving polynomial equations in factored form
- Connecting factors, zeros, and x-intercepts

LEARNING OBJECTIVES

- 1 Multiply polynomials and relate the process to the area model from arithmetic.
- 2 Factor a quadratic expression by an appropriately chosen method.
- 3 Use the zero product property to solve an equation and connect the solutions to the graph.
- 4 Explain why the factored form reveals the zeros while the standard form does not.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Multiplies polynomials at 92% accuracy.
- Factors quadratics at 88% accuracy across all strategies.

- Connects factors, zeros, and intercepts correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The zero product property is applied to a non-zero product: $(x-2)(x+3) = 6$ gives $x-2 = 6$.	The property requires zero on one side. This error is extremely common; check the right-hand side aloud before applying the property, every time.
Signs are mishandled when factoring trinomials with a negative constant term.	Determine the signs before searching for the numbers: a negative constant means the factors have opposite signs. Structure the search rather than guessing.
The GCF is not factored out first, making the remaining factorization much harder.	Always check for a GCF as step one. Make it a written checklist item until it is automatic.

TEACHING NOTE

Factoring is a means, not an end. Every factoring lesson should end by connecting the factored form back to the graph — otherwise the student experiences a term's worth of manipulation with no visible purpose.

UNIT 6 · 14-15 LESSONS

Quadratic Functions and Equations

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TOPICS COVERED

- Quadratic functions as constant second differences
- The parent function $f(x) = x^2$ and its transformations
- Vertex form and the meaning of each parameter
- Standard form, factored form, and vertex form, and what each reveals
- Converting among the three forms
- Axis of symmetry, vertex, intercepts, and end behaviour
- Solving quadratic equations by graphing and by inspection
- Solving by factoring and the zero product property
- Solving by taking square roots
- Completing the square, and deriving vertex form from standard form
- Deriving the quadratic formula by completing the square
- The discriminant and the number and nature of the solutions
- Complex solutions introduced informally as a preview
- Modelling with quadratics: projectile motion, area, and revenue problems
- Comparing linear, exponential, and quadratic models

LEARNING OBJECTIVES

- 1 Convert a quadratic among standard, factored, and vertex form and say what each reveals.
- 2 Solve a quadratic equation by an appropriately chosen method and justify the choice.
- 3 Derive the quadratic formula by completing the square on the general form.

- 4 Use the discriminant to predict the number of real solutions before solving.
- 5 Build and interpret a quadratic model, including the meaning of the vertex in context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves quadratics at 90% accuracy across all four methods.
- Converts among the three forms correctly in 9 of 10 trials.
- Interprets the vertex and intercepts in a modelling context in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The quadratic formula is applied without writing the equation in standard form first.	Set the equation equal to zero before identifying a, b, and c. Require these to be written and labelled separately.
Only the positive square root is taken when solving $x^2 = 16$.	Both roots. Connect to the graph: a horizontal line crosses the parabola twice.
The vertex is reported as a single number rather than as a point.	The vertex is (h, k). Require both coordinates and an interpretation of each in the modelling context.
Signs are dropped in $-b/2a$ when b is negative.	Substitute with parentheses around every value, always. This one habit eliminates most quadratic formula errors.

TEACHING NOTE

Deriving the quadratic formula rather than presenting it is worth two lessons. Students who have derived it recover it when they forget it; students who memorized it are simply lost. It is also many students' first substantial algebraic derivation.

UNIT 7 • 8-9 LESSONS

Statistics and Data Analysis

Վիճակագրություն և տվյալների վերլուծություն

TOPICS COVERED

- Representing univariate data with dot plots, histograms, and box plots
- Comparing centre and spread of two or more data sets
- Standard deviation as a measure of spread
- The effect of outliers on measures of centre and spread
- Two-way frequency tables; joint, marginal, and conditional relative frequencies
- Recognizing possible association in categorical data
- Scatter plots and describing the form, direction, strength, and outliers
- Fitting a linear function to data and using it to make predictions
- The least-squares regression line and its interpretation
- Interpreting the slope and intercept of a regression model in context
- Residuals and residual plots as a check of linearity
- The correlation coefficient and what it does and does not measure
- Correlation versus causation; confounding variables
- The dangers of extrapolation

LEARNING OBJECTIVES

- 1 Interpret the slope and intercept of a regression line in the context of the data.
- 2 Use a residual plot to judge whether a linear model is appropriate.
- 3 Distinguish correlation from causation and propose a plausible confounding variable.
- 4 Compute and interpret conditional relative frequencies from a two-way table.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Interprets regression parameters in context with units in 9 of 10 trials.
- Judges model appropriateness from a residual plot in 4 of 5 trials.
- Produces a defensible correlation-versus-causation argument in 4 of 5 prompts.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A strong correlation is taken as evidence of a causal relationship.	Require an alternative explanation for every claimed association. This is a habit of mind, and it needs repeated practice, not a single warning.
The correlation coefficient is interpreted as the slope.	r measures strength and direction; the slope measures rate of change. Show two data sets with the same r and very different slopes.
Predictions are made far outside the range of the data.	Extrapolation assumes the pattern continues, which the data cannot support. Use an absurd extrapolation to make the point memorable.

TEACHING NOTE

Placing statistics at the end of the course is conventional and often means it is rushed or dropped. At Վարժարան it is assessed like any other unit, because the reasoning it teaches is used by every student for the rest of their life, whether or not they continue in mathematics.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Linear equations, slope, function notation, and negative-number fluency; identifies gaps requiring a Pre-Algebra bridge.
Unit Check	End of each unit • 50 min	Unit mastery criteria plus two extended-response items requiring justification.
Fluency Probe	Every 3rd lesson • 5 min	Equation solving, exponent rules, and factoring; maintained across the year.
Modelling Task	Once per unit from Unit 3	An open-ended problem requiring the student to choose a function family, build a model, and defend it.
Mid-Year Assessment	After Unit 4	Cumulative Units 1–4; weighted toward functions and linear modelling.
End-of-Course Assessment	After Unit 7	All course outcomes, in a format matching state end-of-course examinations; determines Geometry and Algebra II placement.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60–75 min	Full course across two academic years, or one year at 2 lessons/week.
Standard Year	2 lessons/week, 60 min	Full course in approximately 45 weeks — the most common track.
Accelerated	3 lessons/week	Full course in approximately 30 weeks; for students compacting toward AP Calculus.

TRACK	CADENCE	EXPECTED COMPLETION
Support / Co-Seated	2 lessons/week alongside school	Paced to follow the student's school syllabus, with Վարժարան supplying diagnosis and remediation.
Summer Program	4 lessons/week, 8 weeks	Units 5 and 6 — factoring and quadratics — the most common summer request at this level.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

Տնային աշխատանք և ինքնուրույն պարապմունք

Fifty to sixty minutes per lesson: ten to fifteen problems structured in three parts. Part one is current content. Part two is cumulative review drawn from all previous units, which is what prevents the end-of-year collapse that follows blocked practice. Part three is a single extended problem requiring written reasoning, a chosen model, or an interpretation in context. Every equation solved must be checked; every applied answer must carry units and a sentence of interpretation. Students maintain an error log reviewed at each unit boundary.

PROGRESS MILESTONES REPORTED TO PARENTS

Ուսումնական նվաճումներ

These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Justifies each step of equation solving by a named property.
- ◆ Solves linear equations, inequalities, and absolute value statements.
- ◆ Uses function notation fluently and interprets key features of graphs.
- ◆ Computes and interprets average rate of change.
- ◆ Writes and converts linear equations in three forms; solves systems by three methods.
- ◆ Graphs systems of inequalities and interprets feasible regions.
- ◆ Distinguishes linear from exponential growth and interprets model parameters.
- ◆ Works with rational exponents and radical expressions.
- ◆ Multiplies and factors polynomials and connects factors to zeros.
- ◆ Solves quadratic equations by four methods and derives the quadratic formula.
- ◆ Converts among standard, factored, and vertex form and interprets each.
- ◆ Builds and defends models using linear, exponential, and quadratic families.
- ◆ Interprets regression models and distinguishes correlation from causation.
- ◆ Completed Algebra I — ready for Geometry and Algebra II.

MATERIALS AND RESOURCES

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- Graphing calculator (TI-84 family) or an equivalent graphing tool; required from Unit 2 onward.
- Dynamic graphing software for exploring transformations (teacher-shared).
- Algebra tiles, physical or digital, for the polynomials and factoring unit.
- Real data sets for the modelling tasks and the statistics unit.
- Վարժարան Algebra I problem bank — approximately 2,000 items including 150 open-ended modelling tasks.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.